

Attention is a Smoothed Cubic Spline

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Presentation Date: April 24, 2026

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Motivation

- In harmonic analysis and approximation theory, approximate complicated functions by **sums** or linear combinations of simple functions

$$f(x) \approx f_1 + f_2 + \dots + f_d,$$

Also the common approach in econometrics: sieve, OLS, Nadaraya-Watson estimator, etc.

- In modern AI, approximate complicated functions as **compositions** of simple functions

$$F \approx F_1 \circ F_2 \dots \circ F_d$$

- This is hardly new in math

$$X_0 \xrightarrow{F_1} X_1 \xrightarrow{F_2} \dots \xrightarrow{F_d} X_r$$

Motivation

- Assume $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$
- **sum**: parameters scale in md^n .

$$F(\mathbf{x}) = \sum_{j=1}^m \sum_{i_1=1}^d \cdots \sum_{i_n=1}^d \alpha_{j,i_1,\dots,i_n} \left[\prod_{i_k=1}^n \phi_{i_k}(x_k) \right] \mathbf{e}_j$$

- **composition**: parameters scale like $O(mn + n^2)$.

$$F(x) = A_d \sigma_{d-1} A_{d-1} \sigma_{d-2} \cdots \sigma_1 A_1 x$$

where for example A_i are linear transformations and σ_i are nonlinear activations.

- Also, both work well with derivatives due to linearity / chain rule.

Key Results from the Paper

1. With ReLU activation, attention, masked attention, encoder-decoder attention are all cubic splines. Since these components make up the transformer, the transformer is a higher-order spline.
2. Furthermore, the converse is also true by a special case of the Pierce-Birkhoff conjecture, that is, every spline is a ReLU-activated encoder.

Goals today

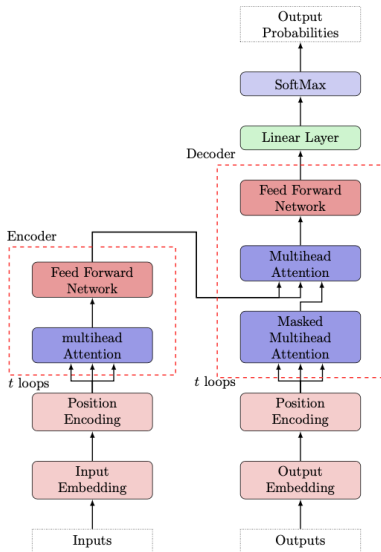
1. Background on Transformers and the attention mechanism.
2. Mathematical Definition of ReLU Transformer and Spline.
3. Proof sketch of the first direction: attention is a cubic spline.
4. Proof sketch of the second direction: every spline is a ReLU-activated encoder.

Background

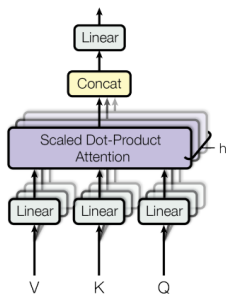
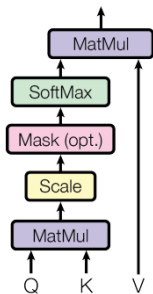
Background on Transformers

- Transformers (Vaswani et al., NeurIPS 2017), underlies many modern AI technologies. Vast majority of LLMs are based on the transformer architecture. Other important applications also, e.g. AlphaFold3, ...
- Core component of the transformer is the attention mechanism.
- In Economics: not yet commonly used in applied settings. So far transformers are mainly used for prediction problems.
- More recently however, elements of ML architecture and activations are being explored in Economics, e.g. ReLU for maximum score (Chen, Gao, and Wen (2025)) or ReLU for inference on welfare functionals (Chen, Chen and Gao (2025)).

Background on Transformers

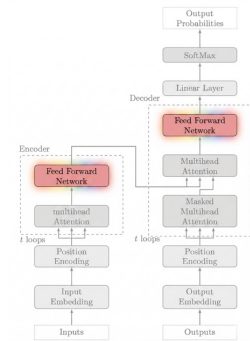


Background on Transformers



Transformer and Spline Definitions

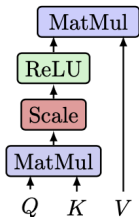
Feed-Forward Neural Network



An l -layer feed-forward neural network is a function $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}^{n_{l+1}}$ given by

$$\varphi(x) = A_{l+1}\sigma_l A_l \sigma_{l-1} \dots \sigma_1 A_1 x + b_{l+1}.$$

Attention & Multihead Attention

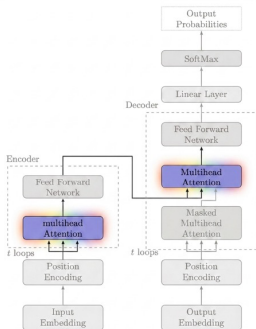


An attention module is a map

$\alpha : \mathbb{R}^{n \times p} \rightarrow \mathbb{R}^{m \times p}$ where

$$\alpha(X) := V(X) \text{ReLU}(K(X)^\top Q(X)).$$

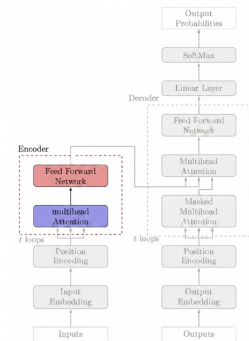
Where Q, K, V are affine linear layers.



A multihead attention module stacks h independent modules:

$$\alpha(X) = (\alpha_1(X), \dots, \alpha_h(X)).$$

Encoder



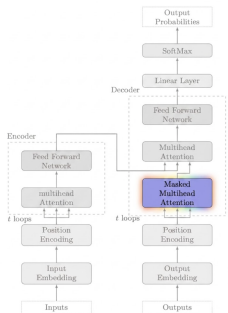
An encoder block composes multihead attention α with a neural network φ :

$$\epsilon = \varphi \circ \alpha.$$

A t -layer encoder chains t blocks

$$\epsilon_t = \varphi_t \circ \alpha_t \circ \dots \circ \varphi_1 \circ \alpha_1.$$

Masked Attention



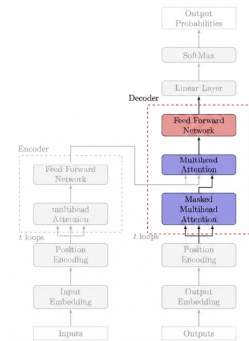
The mask function is

$$\text{mask}(X)_{ij} = \begin{cases} x_{ij} & \text{if } i \leq j, \\ -\infty & \text{if } i > j. \end{cases}$$

The masked attention module is

$$\beta(X) = V(X) \text{ReLU}(\text{mask}(K(X)^{\top} Q(X))).$$

Decoder

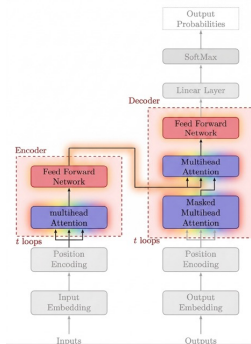


A decoder block is the analogue of an encoder block with masked attention

$$\delta = \varphi \circ \beta.$$

Note: The figure is not quite a decoder block. Instead, it is an encoder-decoder.

Encoder-Decoder Attention



Encoder-decoder attention uses queries from the decoder (Y), but keys and values from the encoder (X):

$$\gamma(X, Y) := V(X) \text{ReLU}(K(X)^\top Q(Y))$$

And, an encoder-decoder block combines masked attention, encoder-decoder attention, and a neural network:

$$\tau(X, Y) = \varphi(\gamma(X, \beta(Y))).$$

Summary of Transformer Components

Component	Mathematical Definition
Feed-Forward Network	$\varphi(x) = A_{l+1}\sigma_l A_l \dots \sigma_1 A_1 x + b_{l+1}$
Attention	$\alpha(X) = V(X)\text{ReLU}(K(X)^\top Q(X))$
Multihead Attention	$\alpha(X) = (\alpha_1(X), \dots, \alpha_h(X))$
Masked Attention	$\beta(X) = V(X)\text{ReLU}(\text{mask}(K(X)^\top Q(X)))$
Encoder-Decoder Attention	$\gamma(X, Y) = V(X)\text{ReLU}(K(X)^\top Q(Y))$
Encoder Block	$\epsilon = \varphi \circ \alpha$
Decoder Block	$\delta = \varphi \circ \beta$
Encoder-Decoder Block	$\tau(X, Y) = \varphi(\gamma(X, \beta(Y)))$

Semialgebraic Partition

Suppose we have b polynomials, consider the mappings $\theta : \{1, \dots, b\} \rightarrow \{1, 0, -1\}$. Any $\pi_1, \dots, \pi_b \in \mathbb{R}[x_1, \dots, x_n]$ induces a particular **sign partition** of $\mathbb{R}^n$

$$\Pi_\theta := \{x \in \mathbb{R}^n : \text{sgn}(\pi_i(x)) = \theta(i), i = 1, \dots, b\}.$$

Now define the entire set of mappings θ of size 3^b ,

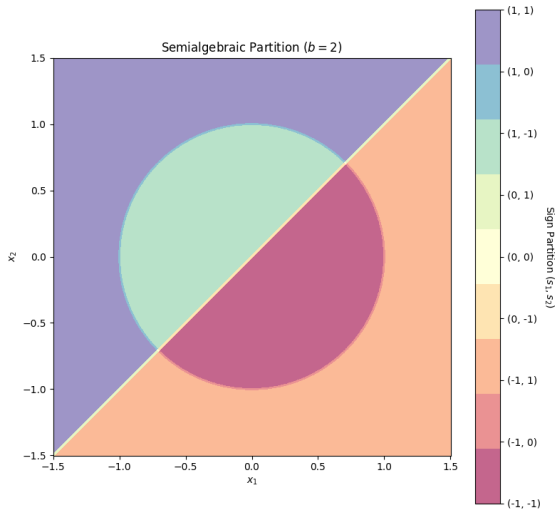
$$\Theta_b := \{\theta : \{1, \dots, b\} \rightarrow \{1, 0, -1\}\}.$$

This is the set of ternary numerals with b bits.

Then $\{\Pi_\theta : \theta \in \Theta_b\}$ is a partition of $\mathbb{R}^n$, the **semialgebraic partition** induced by $\pi_1, \dots, \pi_b$.

Semialgebraic Partition

$\mathbb{R}^2$ with $\pi_1(x_1, x_2) = x_2 - x_1$ and $\pi_2(x_1, x_2) = x_1^2 + x_2^2 - 1$.



Spline

Definition (Spline)

Let $\{\Pi_\theta : \theta \in \Theta_b\}$ be the semialgebraic partition induced by $\pi_1, \dots, \pi_b \in \mathbb{R}[x_1, \dots, x_n]$. A continuous function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a **polynomial spline** of degree k if for each $i = 1, \dots, b$:

1. π_i has degree not more than k ;
2. if $\Pi_\theta \neq \emptyset$, then f restricts to a polynomial of degree not more than k on Π_θ , i.e., $f(x) = \xi_\theta(x)$ for all $x \in \Pi_\theta$, for some $\xi_\theta \in \mathbb{R}[x_1, \dots, x_n]$ of degree not more than k .

Call a degree $\leq k$ spline $S_k(\pi_1, \dots, \pi_b)$. This can be extended to any finite-dimensional real vector space $\mathbb{V}$ by the tensor product $S_k(\pi_1, \dots, \pi_b) \otimes \mathbb{V}$.

Transformers are splines

Past Related Result

Theorem (Arora-Basu-Mianjy-Mukherjee (ICML 2018))

Every neural network $\psi : \mathbb{R}^n \rightarrow \mathbb{R}$ is a linear spline, and every linear spline $l : \mathbb{R}^n \rightarrow \mathbb{R}$ can be represented by a neural network with at most $\lceil \log_2(n+1) \rceil + 1$ depth.

Spline Degrees under Composition and Multiplication

Lemma (Spline composition and multiplication)

(i) Let $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a spline of degree k and $f : \mathbb{R}^m \rightarrow \mathbb{R}^p$ a spline of degree k' . Then $f \circ g$ is a spline of degree kk' .

(ii) Let $f : \mathbb{R}^{r \times s} \rightarrow \mathbb{R}^{m \times n}$ and $g : \mathbb{R}^{r \times s} \rightarrow \mathbb{R}^{n \times p}$ be splines of degrees k and k' . Then $fg : \mathbb{R}^{r \times s} \rightarrow \mathbb{R}^{m \times p}, X \mapsto f(X)g(X)$, is a spline of degree $k + k'$.

Transformer Components as Splines

Theorem 3.3 (Lai, Lim, & Liu; spline.pdf)

- (i) An **attention module** is a cubic spline: $\alpha(X) = V(X)\text{ReLU}(K(X)^\top Q(X))$.
- (ii) A **masked attention module** is a cubic spline:
 $\beta(X) = V(X)\text{ReLU}(\text{mask}(K(X)^\top Q(X)))$.
- (iii) An **encoder-decoder attention module** is a cubic spline:
 $\gamma(X, Y) = V(X)\text{ReLU}(K(X)^\top Q(Y))$.
- (iv) An **encoder block** is a cubic spline: $\epsilon = \varphi \circ \alpha$.
- (v) A **decoder block** is a cubic spline: $\delta = \varphi \circ \beta$.
- (vi) An **encoder-decoder block** is a quintic spline: $\tau(X, Y) = \varphi(\gamma(X, \beta(Y)))$.
- (vii) A **t-layer encoder** is a spline of degree 3^t : $\epsilon_t = \varphi_t \circ \alpha_t \circ \cdots \circ \varphi_1 \circ \alpha_1$.
- (viii) A **t-layer decoder** is a spline of degree 3^t : $\delta_t = \varphi_t \circ \beta_t \circ \cdots \circ \varphi_1 \circ \beta_1$.
- (ix) An **encoder-decoder** with s layers of encoder blocks and t layers of encoder-decoder blocks is a spline of degree $3^{t+s} + 3^t - 3^s$.

Proof sketch of Lemma 3.2

For $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $f : \mathbb{R}^m \rightarrow \mathbb{R}^p$, $f \circ g$ is a spline of degree kk' . For $f : \mathbb{R}^{r \times s} \rightarrow \mathbb{R}^{m \times n}$, $g : \mathbb{R}^{r \times s} \rightarrow \mathbb{R}^{n \times p}$, fg is a spline of degree $k + k'$.

Proof Sketch:

- g partitions $\mathbb{R}^n$ (via polynomials $L = \{\pi_1, \dots, \pi_b\}$). f partitions $\mathbb{R}^m$ (via polynomials $\{\rho_1, \dots, \rho_c\}$). The boundaries cannot be directly “refined” together because they exist in different dimensional spaces.
- Define $\xi_{i,\theta} : \mathbb{R}^n \rightarrow \mathbb{R}^1$ as the polynomial piece of g that is in the partition Π_θ for $i = 1, \dots, m$, and $\zeta_\phi : \mathbb{R}^m \rightarrow \mathbb{R}^p$ as the polynomial piece of f that is in the partition Π_ϕ for $\phi \in \Phi_c$.
- Map f 's boundaries ($\rho_j : \mathbb{R}^m \rightarrow \mathbb{R}$) backwards into $\mathbb{R}^n$ by composing them with g 's interior polynomial pieces ($\xi_{i,\theta} : \mathbb{R}^n \rightarrow \mathbb{R}$). Create the unified partition $M := L \cup \{\rho_j \circ (\xi_{1,\theta}, \dots, \xi_{m,\theta})\}$
- Within any specific partition Π_ϕ of M :

$$f \circ g(x) = \underbrace{\zeta_{\phi|_H}}_{\deg \leq k'} \circ \underbrace{(\xi_{1,\phi|_L}, \dots, \xi_{m,\phi|_L})}_{\deg \leq k}(x)$$

Therefore, the degree is bounded by kk' .

- Second part holds directly since f, g can share refined partition. ■

Connections to Econometrics

ReLU in Recent Econometrics Literature

Maximum Score (Chen, Gao, Wen, 2025)

- Replace discontinuous $\mathbb{I}\{x'\theta > 0\}$ with ReLU compositions:

$$g_{+, \theta, h}(x) := [h(x) - [-x'\theta]_+]_+$$

$$g_{-, \theta, h}(x) := [-h(x) - [x'\theta]_+]_+$$

The authors' ReLU-based maximum score estimator attains convergence rate of $n^{-s/(2s+1)}$ under order- s Hölder smoothness.

Welfare Functionals (Chen, Chen, Gao, 2025)

- Evaluate optimal treatment assignment via the welfare functional

$$W(h_0) := \int [h_0(x)]_+ f(x) dx$$

.

Splines are transformers

Veronese Map

Informally, the degree- k Veronese embedding v_k is the map that takes variables $x_1, \dots, x_n$ to the monomials of degree not more than k in $x_1, \dots, x_n$.

This defines an injective smooth function:

$$v_k : \mathbb{R}^n \rightarrow \mathbb{R}^{\binom{n+k}{k}}, \quad (x_1, \dots, x_n) \mapsto (1, x_1, \dots, x_n, x_1^2, x_1 x_2, \dots, x_n^k).$$

The value $\binom{n+k}{k}$ gives the number of monomials in n variables of degree not more than k .

Simple Examples:

- $v_k : \mathbb{R} \rightarrow \mathbb{R}^{k+1}, \quad v_k(x) = (1, x, x^2, \dots, x^k)$
- $v_2 : \mathbb{R}^2 \rightarrow \mathbb{R}^6, \quad v_2(x, y) = (1, x, y, x^2, xy, y^2)$

Pierce-Birkhoff Conjecture

Conjecture (Pierce-Birkhoff, 1956)

For every spline $f : \mathbb{R}^n \rightarrow \mathbb{R}$, there exists a finite set of polynomials $\xi_{ij} \in \mathbb{R}[x_1, \dots, x_n]$ such that

$$f(x) = \max_{i=1, \dots, m} \min_{j=1, \dots, p} \xi_{ij}(x).$$

Example

Consider the ReLU function $f(x) = \max\{0, x\}$. By setting $m = 2$ and $p = 1$, define:

- $\xi_{11}(x) = 0$
- $\xi_{21}(x) = x$

The Pierce-Birkhoff form correctly yields:

$$f(x) = \max\{\min(0), \min(x)\} = \max\{0, x\}.$$

Equivalence

Lemma 3.5

The Pierce-Birkhoff conjecture holds if and only if for any spline $f : \mathbb{R}^n \rightarrow \mathbb{R}$, there exist $k \in \mathbb{N}$ and a linear spline $l : \mathbb{R}^{\binom{n+k}{k}} \rightarrow \mathbb{R}$ such that $f = l \circ v_k$.

Proof Sketch:

- ($\Rightarrow$) If PB holds, f is a min-max of polynomials ξ_{ij} . Since polynomials of degree $\leq k$ are linear combinations over the Veronese map v_k , applying min-max to these linear combinations yields a linear spline l . Thus, $f = l \circ v_k$.
- ($\Leftarrow$) If $f = l \circ v_k$, we use the fact that any linear spline l can be written as a min-max of linear functions. Composing these linear functions with v_k turns them back into polynomials ξ_{ij} , recovering the PB conjecture form $f = \min_i \max_j \xi_{ij}$. ■

Building the Veronese Map

Lemma 3.6 (Quadratic Veronese as encoders)

Let $v_2 : \mathbb{R}^{n \times p} \rightarrow \mathbb{R}^{(np+2)(np+1)/2}$ be the quadratic Veronese map. There exists a two-layer encoder ϵ_2 such that every column of $\epsilon_2(X)$ contains a copy of $v_2(X)$.

More precisely, this can be constructed by chaining attention modules and one-layer neural networks:

$$\epsilon_2 = \varphi_2 \circ \alpha_2 \circ \varphi_1 \circ \alpha_1$$

Proof Strategy:

- **Layer 1** (α_1, φ_1): Attention α_1 broadcasts every input entry and the constant 1 into every column. NN φ_1 isolates these values to build all linear terms (degree-1 Veronese map) per column.
- **Layer 2** (α_2, φ_2): The attention module α_2 multiplies these linear terms to form partial quadratic terms like $x \text{ReLU}(y)$ and $x \text{ReLU}(-y)$. NN φ_2 subtracts them to recover the exact product $xy = x \text{ReLU}(y) - x \text{ReLU}(-y)$, completing the quadratic Veronese map.

Neural Networks as Encoder Blocks

While the definition of an encoder block $\epsilon = \varphi \circ \alpha$ allows for multilayer neural networks, standard transformer architectures often restrict φ to a single hidden layer.

Lemma 3.7 (1-layer neural networks as encoder blocks)

Let $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}^{n_2}$ be a one-layer neural network. Applying it columnwise to a matrix X :

$$\varphi(X) = [\varphi(x_1), \dots, \varphi(x_p)]$$

is an encoder block of the form $\varphi_1 \circ \alpha_1$, where φ_1 is also a one-layer neural network and α_1 is an attention module.

Proof Strategy:

- **Attention (α_1):** Use a multihead attention module to scatter the input matrix X such that its columns $x_1, \dots, x_p$ are isolated along a large block-diagonal structure.
- **Neural Network (φ_1):** Define an affine map ψ that sums these blocks, acting as a left-inverse where $\psi(\alpha_1(X)) = X$. Because ψ is linear, it can absorb into the original φ to create a new one-layer network $\varphi_1 = \varphi \circ \psi$.

Splines as Encoders

Theorem 3.8

Let $f : \mathbb{R}^{n \times p} \rightarrow \mathbb{R}^{r \times p}$ be a max-definable function. Then f is a t -layer encoder for some finite $t \in \mathbb{N}$ given by:

$$f = \varphi_t \circ \alpha_t \circ \cdots \circ \varphi_1 \circ \alpha_1$$

Furthermore, given the Pierce-Birkhoff conjecture, then any degree- k spline is an encoder.

Proof Strategy:

- Express the max-definable function as a linear spline evaluated on a sufficiently high-degree Veronese map.
- Repeatedly compose the quadratic Veronese encoder (Lemma 3.6) to reach the required degree, then apply the linear spline via the final neural network layer.

Equivalence

Corollary 3.9

If the Pierce-Birkhoff conjecture holds, then the following classes of functions are all equivalent:

- (i) Splines
- (ii) Encoders (Transformers)
- (iii) Max-definable functions
- (iv) Linear splines composed with the Veronese map ($l \circ v_k$)

Proof Sketch:

- Combine Theorem 3.8 (all max-definable functions are encoders) with the Pierce-Birkhoff conjecture (all splines are max-definable functions).
- The remaining equivalences naturally follow from the established fact that neural networks act precisely as linear splines. ■

Some Key Insights

- While neural networks are linear splines, the attention module allows for higher-order complexity.
- Quadratic Veronese maps break out of degree 1 neural networks.
- Layering provides an efficient way to increase model complexity, with the spline degree (3^t) growing exponentially.

Conclusion

- Provided a definition of transformer, and splines in the more general semialgebraic partition setting.
- Established that transformer components are splines, and that every spline can be represented as a transformer under the Pierce-Birkhoff conjecture.
- Relation to Econometrics.

Appendix

Appendix: Some definitions for learning

Definition (Ring)

A ring is a set R equipped with two binary operations: addition $(+)$ and multiplication $(\cdot)$, satisfying the following properties:

- R is an abelian group under addition.
- Multiplication is associative: $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ for all $a, b, c \in R$.
- Multiplication distributes over addition: $a \cdot (b + c) = a \cdot b + a \cdot c$ and $(a + b) \cdot c = a \cdot c + b \cdot c$ for all $a, b, c \in R$.